

Graph & Table

1. Duration Variability (Latency Distributions)

As you noted, this measures how stable the processing time is for different modules, represented as histograms in the paper (labeled under Tables 2b and 3b).

- **X-axis:** Duration (ms).
- **Y-axis:** Probability density.
- **Legends/Labels:** 2 labels representing the frame rate tests: "20 fps" (blue line) and "0.5 fps" (orange line).
- **Structure:** There are separate graphs generated for each VSLAM module (Tracking/VIO Front-End, Local Mapping/VIO Back-End, Loop Closing/RPGO) and split by the device running them (Jetson vs. Laptop).

2. System Capacity (Input Frequency vs. Throughput)

This evaluates whether a module can process data as fast as it arrives. It is represented strictly in data tables (Tables 2, 3, and 4) rather than plots.

- **Columns:** Input Freq., Throughput, and Duration (showing the mean $\pm$ standard deviation).
- **Rows:** The specific VSLAM modules (Tracking, Local Mapping, Loop Closing/GBA).
- **Structure:** The tables are subdivided into quadrants comparing the edge hardware (Jetson) versus the high-end hardware (Laptop) at both 20 fps and 0.5 fps.

3. Degree of Concurrency (CPU Usage and Outcomes)

This evaluates how hard the hardware is working and the penalty of that work, shown as a three-part stacked graphic in Figure 5.

- **Top Graphs (CPU Usage Over Time):**
 - **X-axis:** Time (s).
 - **Y-axis:** CPU Usage (%).
 - **Legends:** Device type (Laptop, Jetson) and Frame Rates (20 fps, 15 fps, 10 fps, 5 fps).
- **Middle Graphs (Average CPU Usage):**
 - **X-axis:** Frame Rate (fps).
 - **Y-axis:** CPU Usage (%) shown as bar charts with error bars.
- **Bottom Graphs (Frame/KeyFrame Outcomes):**
 - **X-axis:** Device and Frame Rate combinations (e.g., L 20, J 20, L 15).
 - **Y-axis:** Absolute count (from 0 up to 3000).
 - **Legends (4 labels):** Tracked F (solid blue), Dropped F (hashed blue), Created KF (solid orange), Dropped KF (hashed orange).

4. Map Complexity (Covisibility Graph Density)

This evaluates the structural connections of the map to see how data is shared, shown as node-link diagrams rather than standard plots in Figure 6.

- **Nodes:** Represented as orange dots, which indicate individual KeyFrames.
- **Edges:** Represented as gray/black lines connecting nodes. An edge exists if two KeyFrames share visible MapPoints/Landmarks.
- **Structure:** Shows visual comparisons between algorithms (ORB-SLAM3's dense global map vs. Kimera's sparser, sliding-window map) and sequence difficulty (easy vs. hard).

5. Hardware Stress Testing (Performance Differences)

This isolates hardware limitations from algorithmic limitations across the entire dataset, shown as Heatmaps in Figures 3 and 4.

- **X-axis:** The specific EuRoC dataset sequence being run (MH01, MH02, ..., V203).
- **Y-axis:** The Frame Rate (20, 15, 10, 5).
- **Cell Values/Colors:** The numerical difference between the Jetson value minus the Laptop value, mapped to a purple (positive) to brown (negative) color scale.
- **Sub-charts:** Divided into 4 metrics: ATE (m), Diff Frames Processed (%), Diff KeyFrames Dropped (%), and Diff KeyFrames Created (%).